The Journal of Open Source Software

# CUDA-METRO: Parallel Metropolis Monte-Carlo for 2D Atomistic Spin Texture Simulation

**Arkavo Hait** [1] **and Santanu Mahapatra** [1]

**1** Nano-Scale Device Research Laboratory, Department of Electronic Systems Engineering, Indian Institute of Science (IISc) Bangalore, Bangalore 560012, India.

**DOI:** 10.xxxxxx/draft

**Software**

- Review
- Repository
- Archive

**Editor:**

**Submitted:** 17 October 2024
**Published:** unpublished

**License**
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

## Statement of need

Atomistic spin texture simulations are crucial for understanding and predicting the behaviour of magnetic materials at the nanoscale. These simulations provide insights into fundamental properties like magnetic phase transition and are thus useful for exploring novel materials (Kabiraj & Mahapatra, 2023). The Metropolis(Metropolis et al., 1953) Monte-Carlo(Heisenberg, 1928) (MC) method is frequently utilised for atomistic spin texture simulations as a sampling algorithm to investigate the phase space of a system and is especially effective for calculating equilibrium properties (Evans et al., 2014; Müller et al., 2019). Efficient parallelization of Metropolis Monte Carlo simulation is challenging since the evolving states are typically not independent because of the Markov property. Here we focus on simulating magnetic phase transition under the anisotropic Heisenberg Model in a very high dimensional space, which is important for emerging two-dimensional (2D) magnetism and nontrivial topological spin textures (Augustin et al., 2021). Previous attempts for parallelization are restricted to the simpler Ising Model and not applicable to 2D materials because of their finite magneto crystalline anisotropy, complex crystal structures and long-range interactions. MC simulation of the anisotropic Heisenberg model is very complex owing to the large number of additional Hamiltonian calculations and interconnectivity between lattice points. The amount of calculations increases as $^{2}$, where represents the dimension of a square lattice. This becomes alarming when exceeds 100, which is entirely justifiable for investigating topological spin textures (skyrmions, merons, etc.) Here we present CUDA-METRO, a graphical processing unit (GPU) based open source code for accelerated atomistic spin dynamics simulation. We evaluated our code by precisely simulating complex topological spin textures and temperature-dependent magnetic phase transitions for diverse 2D crystal structures with long-range magnetic interactions. We demonstrate exceptional acceleration while finding the ground state of a $750 \times 750$ supercell in 9 hours using an A100-SXM4 GPU.

## Summary

We consider a lattice system with a periodic arrangement of atoms, where each atom is represented by a 3D spin vector. This atomistic spin model is founded on the spin Hamiltonian, which delineates the essential spin-dependent interactions at the atomic scale, excluding the influences of potential and kinetic energy and electron correlations. The spin Hamiltonian of the atom is conventionally articulated as

$$= \quad 1 \quad 1 \quad 1 \quad 1 \quad 2 \quad 2$$

$$2 \quad 2 \quad 3 \quad 3 \quad 3 \quad 3$$

$$4 \quad 4 \quad 4 \quad 4 \quad ( \quad )^{2}$$

**Algorithm 1** Metropolis Selection

```
procedure M(, )
    if ¡ 0 then
        Return 1 (ACCEPT)
    else if ¡ then                                  is uniformly random
        Return 1 (ACCEPT)
    else
        Return 0 (REJECT)
```

$$( \text{»} )$$

Where  is the isotropic exchange parameter, the s are the anisotropic exchange parameters, with the superscript denoting the spin direction,  is the single ion exchange parameter,  is the biquadratic parameter,  is the Dyzaloshinskii-Moriya Interaction(DMI) parameter.  is the dipole moment of a single atom and  is the external magnetic field. , are individual atomic spin vectors. $-''$ are the first set of neighbours, $-''$ are the second set of neighbours and so on. The subscripts below all s and s denote the neighbour set, $_1$ denotes the first neighbours, $_2$ the second and so on. From this equation, we see that energy of an atom depends on its interactions with the neighbours. In our code, we have limited the number of neighbour sets to be 4 since it is expected for 2D materials that the interaction energy dies down beyond that. All these above parameters except  are material specific parameters that are the inputs to our MC code.

Starting from a random spin configuration, in this many-body problem, our objective is to find the orientation of spin vectors for every atom so that the energy of the entire lattice reaches to its minimum for a given magnetic field and temperature. Traditionally single spin update(SSU) scheme is employed to solve this problem, which satisfies the detailed balance condition. In the SSU method of updating the state, a single atomic spin is chosen at random and changed, while noting down the energy shift. This new state is then accepted or rejected using the Metropolis criteria as shown in Algorithm 1, where $= ()^1$,  being the Boltzmann constant and  being the temperature. It is imperative that SSU becomes extremely inefficient as the dimensionality increases. Here we propose a parallel algorithm to find the ground state, which is inaccessible by SSU scheme.

In our method, as depicted in Algorithm 2, we select multiple atomic spins at the same time and change them all at once, treating them as independent events. For any individual spin, they do not feel the effects of the other changed spins. In each of these points, we use the Metropolis criteria to accept or reject the changed spin vectors. This becomes our new state. Here  denotes the number of lattice points we are evaluating at the same time for any given state, while  is the batch size. Tuning  ensures that we can fill up our VRAM with pre-generated random numbers instead of generating $4$ »  numbers per step. These 4 random number arrays are further processed into , our site selection, $(,)$, which become the angle coordinates for a new random spin vector and  which is a conditional uniform random number used to evaluate the Metropolis criteria.

At present, five different lattice types (square, rectangular, centred-rectangular, hexagonal and honeycomb) are implemented in our code since most of the 2D magnetic materials fall into this category (Kabiraj et al., 2022), and for neighbour mapping, we use analytical relations (Koziol, 2020).

For a lattice of size » , $100\%$ parallelization would correspond to selecting $^2$ spins at random. Since each spin selection and its consequent Metropolis criterion is evaluated on a separate CUDA core, it becomes apparent that we would need $^2$ CUDA cores to achieve this $100\%$ parallelization.Since the proposed algorithm may not adhere to the detailed balance conditions, it yields approximate results, and there is a trade-off between parallelization/acceleration and

DRAFT

**Algorithm 2** Parallel Monte Carlo

```
procedure STEP(Run)
    Read Initial state
    Create 4 » length uniform random arrays
    Process the 4 uniform random number arrays
    for ¡ do
        Slice the 4 arrays into sub-arrays with range [ »  » ( + 1) 1]
        Execute P parallel BLOCKS with these sub arrays
        for In each BLOCK do
            Evaluate before(0) and after(1) spin change                Multithreading
            Wait for all BLOCKS to finish then increment
        Update all spins to next state
        This state is now our next state
```

accuracy. It is found that if the parallelization is limited to $10\%$ of the lattice size, we obtain very accurate results with significant acceleration.

We define our Monte Carlo Step(MCS) to be a collection of many atomic steps. The exact number depends on the parallelization used in the simulation. Unlike the SSU scheme, we do not extract simulation data after every $^{2}$ spin change, rather we let it run till all the CUDA cores have sychronized and arrived at the next block check, which defines our step. To directly compare, if the SSU has $1$ spins changed between each step, our algorithm will have , where is the number of CUDA cores used or parallelization used. In the conventional SSU, data is extracted at every $^{2}$(a full sweep) steps, compared to which, we would take $\frac{^{2}}{}$ steps to reach the same data point.

We validate our algorithm by accurately reproducing temperature-dependent magnetic phase transitions and intricate topological spin textures already reported in the literature, either by SSU based Monte Carlo or by Landau Lifshitz Gilbert(LLG)(Gilbert, 2004) spin dynamics, for diverse crystal structures.

It was once thought that long-range magnetic order could not exist (Mermin & Wagner, 1966) in two or one-dimensional materials. In 2017, $_{3}$ was one of the first 2D materials (Huang et al., 2017) where ferromagnetism was demonstrated. Consequently, further studies have shown the existence of long-range magnetic order in several other 2D materials. Using CUDA-METRO, we first simulate the magnetic phase transition of $_{3}$ from ferromagnetic to paramagnetic with increasing temperature. The rise of temperature causes the atomic spins not to be locked into a ferromagnetic position but rather gives them enough energy to orient themselves in any direction. The average direction of all the spins would then lead to a null vector, which gives rise to paramagnetism. In Fig 1, we reproduce magnetic phase transition from (Kartsev et al., 2020), and show the effect of parallelization with respect to the reference data. As mentioned before, we can obtain very accurate results with respect to SSU methods by limiting the parallelization at $10\%$ or below of $^{2}$. The ferromagnetic to paramagnetic transition point is known as Critical temperature and is extracted from the peak of the susceptibility versus temperature plot.

Next, we demonstrate the nucleation of topological spin textures which are emergent phenomena in condensed matter physics and are rapidly having importance in information technology(Bessarab et al., 2018; Luo & You, 2021). While MC simulations of medium-sized supercells $(64 \text{ » } 64)$ yield quite accurate results for Critical temperature calculation, observing topological spin textures, once needs much larger supercells.

First, we report the meron and anti-meron, which are structures with topological numbers $1/2$ and $+1/2$ respectively observed in $_{3}$ by LLG equations of spin dynamics. The results of our simulation are shown in Fig 2. $_{3}$ has a honeycomb lattice structure and for this simulation, we have considered up to the third nearest neighbour with biquadratic exchange. We can

DRAFT

see all 4 types of meron and anti-meron structures as found in the previous report(Augustin et al., 2021). This simulation was conducted in a $500 \times 500 (143 \times 143^2)$ supercell and took 300s to stabilize these topological spin textures at a parallelization of $3\%$ conducted on an A100-SXM4 processor.

Second, we simulate skyrmions in $_2$(Cui et al., 2022) as shown in Fig 3. $_2$ is a square lattice and for this simulation, we have considered up to the second nearest neighbour. The uniqueness of this material is that it exhibits anisotropic DMI with an anti-ferromagnetic ground state. Such an anti-ferromagnetic skyrmion spin texture is accurately reproduced in our simulation. Anti-ferromagnetic skyrmions are technologically important since they do not exhibit skyrmion Hall effect. We further study the material $_3$ (Du et al., 2022) which has a hexagonal lattice. This simulation was conducted considering only the nearest neighbours and the formation of skyrmions is shown in Fig 3. Once again our results are in agreement with the original report. All these simulations were conducted in a $200 \times 200 (49 \times 49^2)$ supercell and took 30s to stabilize these topological spin textures at a parallelization of $20\%$ conducted on a V100-SXM2 processor.

In Fig 4 we demonstrate the skyrmion neucleation process for the material (Liang et al., 2020), which has a hexagonal lattice. While we first observe several skyrmions, with evolving MCS, they disappear and the whole lattice eventually becomes uniformly ferromagnetic,which happens to be the direction of the applied magnetic field. This has not been reported in the original literature(Liang et al., 2020), possibly because of the high computational time required for a traditional SSU scheme. In Fig 5, we further show a similar life cycle evolution for a giant skyrmion of diameter $21$ hosted in the material $_{33}$ (Kabiraj & Mahapatra, 2023). To host such a large skyrmion, the simulation was conducted in a supercell of size $750 \times 750$ with a parallelization ratio of $1\%$ utilizing $70\%$ VRAM of an A100-SXM4 GPU. As mentioned before, our parallelization is limited by the number of CUDA cores and so we cannot go more than $1\%$ parallelization for this simulation. However, even with this low parallelization ratio, we can still access 8000 lattice points simultaneously and by careful tuning of our parameter , we can observe the ground state of a $750 \times 750$ supercell in $9$ hours using an A100-SXM4 GPU. The formation of the skyrmion roughly takes $100$ mins.

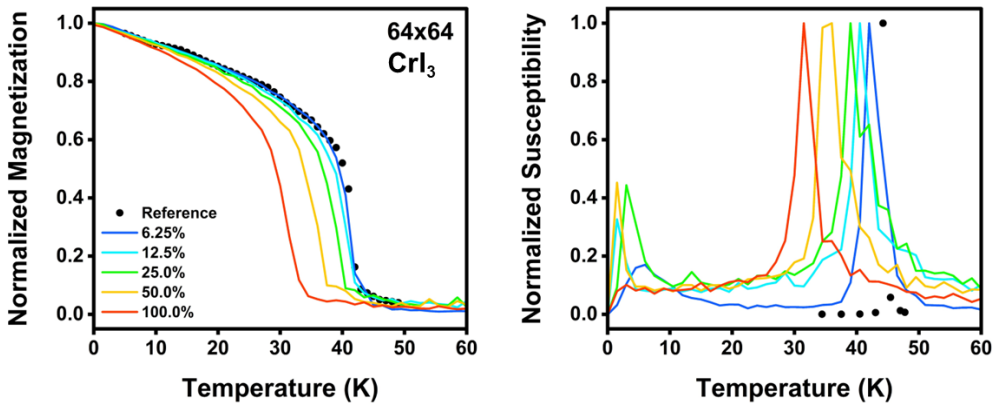


Fig 1: Discrepancy between simulation and reference(Kartsev et al., 2020) results at differing levels of parallelization. At $10\%$, the simulation results are almost indistinguishable from the reference data.

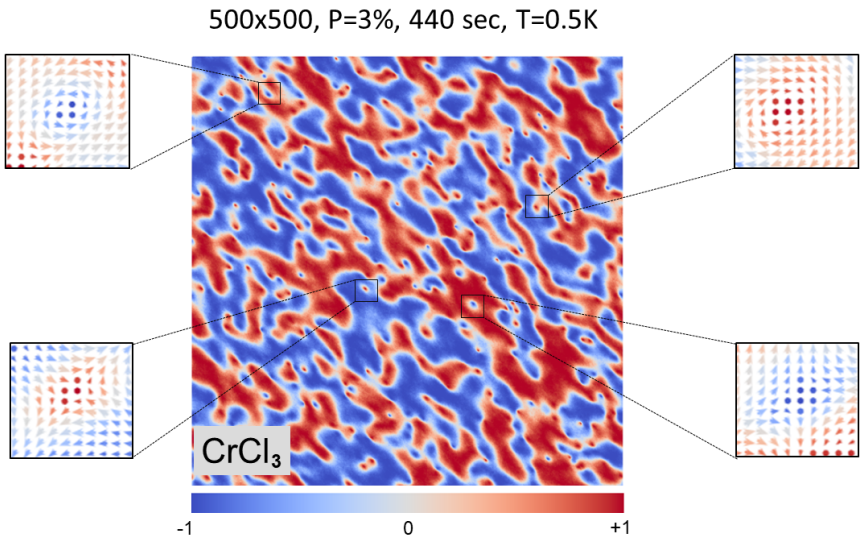


Fig 2: Presence of anti-merons and merons in $_3$. The material parameters are taken from (Augustin et al., 2021). The color bar represents normalized spin vectors in the z direction.

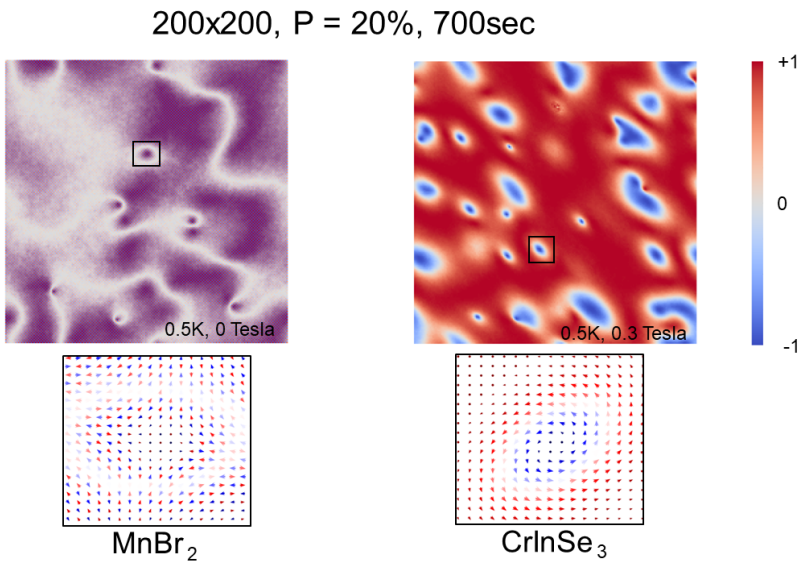


Fig 3: Presence of skyrmions in $_2$ and $_3$. The material parameters are taken from(Cui et al., 2022) and (Du et al., 2022) respectively. The color bar represents normalized spin vectors in the z direction. Note that the spins of $_2$ appear purple because there are red-blue spin pairs for the vast majority.

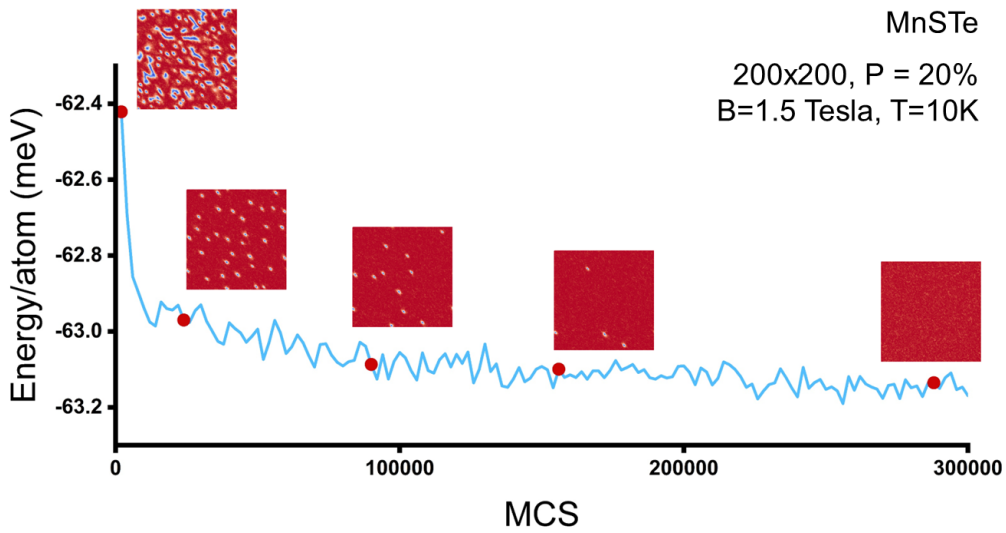


Fig 4: Lifetime of a skyrmion in , from its creation to annihilation. The material parameters are taken from (Liang et al., 2020). The graph denotes the average energy per atom. As we approach the global minima, the entire field becomes aligned to the magnetic field as expected. Total time: $30$.

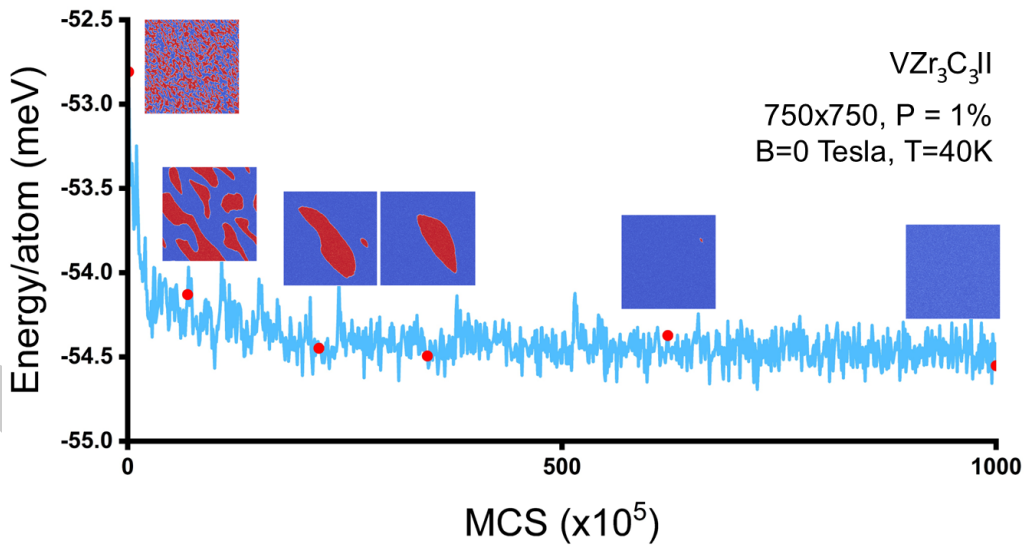


Fig 5: Lifetime of a skyrmion in $_{33}$, from its creation to annihilation. The material parameters are taken from (Kabiraj & Mahapatra, 2023). The graph denotes the average energy per atom. Note how the entire field is now blue (as opposed to red as in Fig 4), this is because unlike the simulation in Fig 4, there is no external magnetic field applied, this means that the ground state would either be all spins up(red) or all spins down(blue) with a $50\%$ probability for either. Total time: $9$ hrs.


## Acknowledgements

This work is supported by the Core Research Grant (CRG) scheme of the Science and Engineering Research Board (SERB), Government of India, under Grant No. CRG/2020/000758.

# References


Augustin, M., Jenkins, S., Evans, R. F. L., Novoselov, K. S., & Santos, E. J. G. (2021). Properties and dynamics of meron topological spin textures in the two-dimensional magnet CrCl3. *Nature Communications*, *12*(1), 185. https://doi.org/10.1038/s41467-020-20497-2

Bessarab, P. F., Müller, G. P., Lobanov, I. S., Rybakov, F. N., Kiselev, N. S., Jónsson, H., Uzdin, V. M., Blügel, S., Bergqvist, L., & Delin, A. (2018). Lifetime of racetrack skyrmions. *Scientific Reports*, *8*(1), 3433. https://doi.org/10.1038/s41598-018-21623-3

Cui, Q., Zhu, Y., Ga, Y., Liang, J., Li, P., Yu, D., Cui, P., & Yang, H. (2022). Anisotropic dzyaloshinskiimoriya interaction and topological magnetism in two-dimensional magnets protected by P4m2 crystal symmetry. *Nano Letters*, *22*(6), 23342341. https://doi.org/10.1021/acs.nanolett.1c04803

Du, W., Dou, K., He, Z., Dai, Y., Huang, B., & Ma, Y. (2022). Spontaneous magnetic skyrmions in single-layer $_3$ (X = Te, Se). *Nano Letters*, *22*(8), 34403446. https://doi.org/10.1021/acs.nanolett.2c00836

Evans, R. F. L., Fan, W. J., Chureemart, P., Ostler, T. A., Ellis, M. O. A., & Chantrell, R. W. (2014). Atomistic spin model simulations of magnetic nanomaterials. *Journal of Physics: Condensed Matter*, *26*(10), 103202. https://doi.org/10.1088/0953-8984/26/10/103202

Gilbert, T. L. (2004). Classics in Magnetics A Phenomenological Theory of Damping in Ferromagnetic Materials. *IEEE Transactions on Magnetics*, *40*(6), 34433449. https://doi.org/10.1109/TMAG.2004.836740

Heisenberg, W. (1928). Zur Theorie des Ferromagnetismus. *Zeitschrift Für Physik*, *49*(9), 619636. https://doi.org/10.1007/BF01328601

Huang, B., Clark, G., Navarro-Moratalla, E., Klein, D. R., Cheng, R., Seyler, K. L., Zhong, D., Schmidgall, E., McGuire, M. A., Cobden, D. H., Yao, W., Xiao, D., Jarillo-Herrero, P., & Xu, X. (2017). Layer-dependent ferromagnetism in a van der waals crystal down to the monolayer limit. *Nature*, *546*(7657), 270273. https://doi.org/10.1038/nature22391

Kabiraj, A., Jain, T., & Mahapatra, S. (2022). Massive Monte Carlo simulations-guided interpretable learning of two-dimensional Curie temperature. *Patterns*, *3*(12), 100625. https://doi.org/10.1016/j.patter.2022.100625

Kabiraj, A., & Mahapatra, S. (2023). Realizing unipolar and bipolar intrinsic skyrmions in MXenes from high-fidelity first-principles calculations. *Npj Computational Materials*, *9*(1), 112. https://doi.org/10.1038/s41524-023-01129-x

Kartsev, A., Augustin, M., Evans, R. F. L., Novoselov, K. S., & Santos, E. J. G. (2020). Biquadratic exchange interactions in two-dimensional magnets. *Npj Computational Materials*, *6*(1). https://doi.org/10.1038/s41524-020-00416-1

Koziol, Z. (2020). Number of equidistant neighbors on honeycomb lattice. *J. Phys. Commun.*, *4*(7), 075003.

Liang, J., Wang, W., Du, H., Hallal, A., Garcia, K., Chshiev, M., Fert, A., & Yang, H. (2020). Very large Dzyaloshinskii-Moriya interaction in two-dimensional Janus manganese dichalcogenides and its application to realize skyrmion states. *Physical Review B*, *101*(18), 184401. https://doi.org/10.1103/PhysRevB.101.184401

Luo, S., & You, L. (2021). Skyrmion devices for memory and logic applications. *APL Materials*, *9*(5), 050901. https://doi.org/10.1063/5.0042917

Mermin, N. D., & Wagner, H. (1966). Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic heisenberg models. *Phys. Rev. Lett.*, *17*, 11331136. https://doi.org/10.1103/PhysRevLett.17.1133


DRAFT

Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., & Teller, E. (1953). Equation of State Calculations by Fast Computing Machines. *The Journal of Chemical Physics*, *21*(6), 10871092. https://doi.org/10.1063/1.1699114

Müller, G. P., Hoffmann, M., DiSSelkamp, C., Schürhoff, D., Mavros, S., Sallermann, M., Kiselev, N. S., Jónsson, H., & Blügel, S. (2019). Spirit: Multifunctional framework for atomistic spin simulations. *Phys. Rev. B*, *99*, 224414. https://doi.org/10.1103/PhysRevB.99.224414

DRAFT